

[PAY-3811] ACH Token Vault never publishes deactivation events — reserved CloudEvent attribute fails every publish, masked by a false-success log

Created: 10/Jun/26 Updated: 08/Jul/26

Status:	In Stage
Project:	Payments
Components:	None
Affects versions:	None
Fix versions:	R9.6

Type:	Bug	Priority:	3: Standard
Reporter:	Jason Thomas	Assignee:	Jay Patel
Resolution:	Unresolved		
Labels:	account-updater, ach-token-vault		
Σ Remaining Estimate:	0 minutes	Remaining Estimate:	0 minutes
Σ Time Spent:	3 hours, 30 minutes	Time Spent:	3 hours, 30 minutes
Σ Original Estimate:	Not Specified	Original estimate:	Not Specified

Issue links:	Blocks blocks PAY-3790 Account Updater orchestrator: NOC cor... Closed				
Sub-tasks:	Key	Summary	Type	Status	Assignee
	PAY-4011	QA: Create Test Cases	Sub-task	To Do	Christopher Ariel Valenzuela Medina
	PAY-4012	QA: Test Fix	Sub-task	To Do	Christopher Ariel Valenzuela Medina
Story Points:	3				
Sprint:	PAY-Sprint 17				
QA Required:	Yes				
Grooming Status:	8-Ad-Hoc				
Pod:	PAY				

Description

Observed Behavior:
NocBankAccountDeactivatedPublisher and AchReturnBankAccountDeactivatedPublisher pass ["subject"] as a CloudEvent extension attribute. Azure.Messaging rejects subject as a reserved attribute name, so every publish of Account.BankToken.NocDeactivated / Account.BankToken.ReturnDeactivated throws — and the vault's framework version (Wellfit.EventGrid 3.0.4) swallows the exception inside PublishAsyncDomain and then logs "Successfully published...". The events never reach Event Grid; the logs say they did.

Expected:
Both deactivation events publish to the EG Domain accounts topic (subject carried via the publish overload's subject parameter), and a failed publish surfaces as an error — never a success log.

Steps to Reproduce:
Local compose e2e (2026-06-10): drive a NOC-deactivation through the vault → log shows Failed to publish event Account.BankToken.NocDeactivated ... ArgumentException ... reserved attribute: 'subject' immediately followed by Successfully published.... Reproducibility: Always.

Impact:
Latent in every environment today (the only consumer — the Account Updater orchestrator — isn't deployed yet). Becomes user-visible at orchestrator promotion: OnAccountDeactivated never fires → AccountUpdater.ReturnDeactivations stays empty → ach-reporting FR-10 (Bank Account Deactivation report) renders blank and [PAY-3790](#) CLOSED AC-2 cannot pass.

AC:

- AC-1: Both vault publishers emit valid CloudEvents — subject via the subject parameter, no reserved extension attributes — with unit tests pinning the call shape (no-reserved-attrs assert).
- AC-2: A vault deactivation in Dev is observable on the accounts SB topic and recorded by the orchestrator (ReturnDeactivations row / NOC resolution).
- AC-3: A failed publish in the vault logs as an **error**, not success — framework exception-swallow raised separately against wellfit-framework-core (linked); vault-side interim mitigation acceptable.

Notes:

- Same defect family as the orchestrator-side publish fix already on the account-updater feature branch; fix shape is identical.
- Framework 3.0.4 PublishAsyncDomain swallow/false-success is its own wellfit-framework-core issue (same family as the TokenRequestHandler cache-key bug) — this ticket only covers the vault's publishers.

Comments

Comment by [Jay Patel](#) [08/Jul/26]

QA Test Plan

Fix merged to main on 2026-06-25 (wellfit-payments PR #290, commit f7e4af2e).

What changed

When a bank account is deactivated in the ACH Token Vault, it must publish an event so the Account Updater orchestrator records it (and the ach-reporting FR-10 *Bank Account Deactivation* report shows it). Previously the vault put the event's routing "subject" in a slot the messaging library rejects, so **every deactivation event silently failed to send** while the logs falsely said "Published...". The fix sends the subject correctly, so the events now actually flow.

Two events are affected, both published to the Event Grid Domain accounts topic; consumer is the OnAccountDeactivated function in the payments function app:

- Account.BankToken.NocDeactivated — a NOC deactivates a bank token
- Account.BankToken.ReturnDeactivated — an ACH return deactivates a bank token

Preconditions (confirm before testing)

1. The ACH Token Vault build deployed to the test env includes PR #290 (merge commit f7e4af2e or later on main).
2. The Account Updater orchestrator (PAY-3790: the OnAccountDeactivated function + [AccountUpdater].[ReturnDeactivations] table) is deployed in the **same** environment — it is the consumer; without it you can only verify the publish side, not end-to-end.
3. Event Grid subscription for the accounts topic → OnAccountDeactivated exists (Dev-confirmed 2026-06-24: dev-evgd-payments/accounts → payments-func-account-deactivated → dev-func-payments/OnAccountDeactivated).

Test cases

TC1 — ACH return deactivation (happy path)

- Do: Process an ACH return for a tokenized bank account using a **hard** return code that deactivates the token — e.g. R02 (Account Closed) or R03 (No Account). Do NOT use R01/R09 (see TC3).
- Expect: Vault logs Published AchReturnBankAccountDeactivatedEvent for TokenId: ...; event lands on the accounts topic and OnAccountDeactivated fires; a new row in [AccountUpdater].[ReturnDeactivations] for that TokenId/SourceEventId; no ArgumentException ... reserved attribute: 'subject' in logs.

TC2 — NOC deactivation (happy path)

- Do: Process a NOC for a tokenized account that results in deactivation (a non-correctable / name-disable NOC — the token ends up Inactive).
- Expect: Vault logs Published AchNocBankAccountDeactivatedEvent for TokenId: ...; OnAccountDeactivated fires; the Applied NOC record for that token is resolved as **Deactivated** (this feeds ach-reporting FR-10, separating NOC-driven deactivations from plain applies); no reserved-attribute exception.

TC3 — Soft return (negative / must NOT fire)

- Do: Process an ACH return with a **soft** return code: R01 (Insufficient Funds) or R09 (Uncollected Funds).
- Expect: Token stays active; no deactivation event; no ReturnDeactivations row. Confirms the fix did not cause over-firing.

TC4 — Log integrity

- Do: During TC1/TC2, inspect App Insights / service logs.
- Expect: The pre-fix signature Failed to publish event Account.BankToken.**Deactivated** ... **reserved attribute: 'subject' is *absent**; success logs now correspond to events that actually reached the consumer.

TC5 — Idempotency (optional)

- Do: Redeliver the same deactivation event / reprocess the same source.
- Expect: Exactly one record (NOC resolved by Applied-state; return deduped by SourceEventId) — no duplicates.

Out of scope — do NOT test

- **AC-3 "a failed publish must log as an error, not success."** Not fixed here — it is a known framework limitation (`Wellfit.EventGrid` swallows publish failures fleet-wide) tracked separately in `wellfit-framework-core` for the architect. A publish that fails may still log success; that is expected for this ticket.

Pass criteria

TC1 + TC2 produce the correct downstream records with no reserved-attribute exceptions; TC3 correctly produces nothing. End-to-end this satisfies AC-1 (valid events, unit-test-pinned) and AC-2 (vault deactivation observable → orchestrator records).

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